

AA/ME 507 Winter 2026

Homework 5

March 6, 2026

Due Date: March 15, 2026 (11:59 PM)

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Problem 1. Similarity solutions: Lamb–Oseen vortex An incompressible viscous fluid occupies the entire plane. Assume the motion is axisymmetric and purely azimuthal, so that in cylindrical coordinates (r, θ, z) ,

$$u_r = 0, \quad u_z = 0, \quad u_\theta = u_\theta(r, t).$$

Let $\omega(r, t)$ denote the axial vorticity. Suppose the total circulation is fixed:

$$\Gamma = \int_0^\infty \omega(r, t) 2\pi r \, dr.$$

1. Starting from the Navier–Stokes equations, show that $\omega(r, t)$ satisfies

$$\frac{\partial \omega}{\partial t} = \nu \left(\frac{\partial^2 \omega}{\partial r^2} + \frac{1}{r} \frac{\partial \omega}{\partial r} \right).$$

2. Show by dimensional analysis that a similarity solution may be sought in the form

$$\omega(r, t) = \frac{\Gamma}{\nu t} f(\eta), \quad \eta = \frac{r}{\sqrt{\nu t}}.$$

3. Derive the ordinary differential equation satisfied by $f(\eta)$.
4. Try solutions of the form $f(\eta) = Ae^{-\eta^2/4}$ and use the circulation constraint to determine A .
5. Hence show that

$$\omega(r, t) = \frac{\Gamma}{4\pi\nu t} \exp\left(-\frac{r^2}{4\nu t}\right).$$

6. Using the relation between azimuthal velocity and vorticity, derive

$$u_\theta(r, t) = \frac{\Gamma}{2\pi r} \left(1 - \exp\left[-\frac{r^2}{4\nu t}\right] \right).$$

7. Find the radius at which u_θ is maximum, and show how that radius scales with time.

Problem 2. Separation of variables: start-up Poiseuille flow A viscous incompressible fluid of kinematic viscosity ν and dynamic viscosity μ is confined between two infinite stationary parallel plates located at $y = -h$ and $y = h$. For $t < 0$, the fluid is at rest. At $t = 0$, a constant pressure gradient

$$-\frac{\partial p}{\partial x} = G, \quad G > 0,$$

is suddenly imposed and maintained for all $t > 0$. Assume the flow is unidirectional,

$$\mathbf{u} = (u(y, t), 0, 0),$$

and that u depends only on y and t .

1. Starting from the Navier–Stokes equations, show that $u(y, t)$ satisfies

$$\frac{\partial u}{\partial t} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \frac{\partial^2 u}{\partial y^2}, \quad -h < y < h, \quad t > 0.$$

2. Write the boundary and initial conditions:

$$u(-h, t) = 0, \quad u(h, t) = 0, \quad u(y, 0) = 0.$$

3. Show that the steady solution is

$$u_s(y) = \frac{G}{2\mu} (h^2 - y^2).$$

4. Define the transient part

$$v(y, t) = u(y, t) - u_s(y),$$

and show that $v(y, t)$ satisfies

$$\frac{\partial v}{\partial t} = \nu \frac{\partial^2 v}{\partial y^2},$$

with homogeneous boundary conditions

$$v(-h, t) = 0, \quad v(h, t) = 0,$$

and initial condition

$$v(y, 0) = -\frac{G}{2\mu} (h^2 - y^2).$$

5. Use separation of variables in the form

$$v(y, t) = Y(y)T(t)$$

to derive the eigenvalue problem for $Y(y)$, and show that the eigenfunctions may be written as

$$Y_n(y) = \sin\left(\frac{n\pi(y+h)}{2h}\right), \quad n = 1, 2, 3, \dots$$

or, equivalently, in a cosine basis adapted to the symmetry of the problem.

6. Determine the coefficients in the series solution and obtain the velocity field $u(y, t)$ for all $-h < y < h$ and $t > 0$.

7. Show that

$$u(y, t) \rightarrow \frac{G}{2\mu} (h^2 - y^2) \quad \text{as } t \rightarrow \infty,$$

and determine the characteristic diffusive time scale for the approach to steady state.

8. Plot the centerline velocity $u(0, t)$ versus time when retaining 1 and 10 modes in the series.

Problem 3. Lubrication theory: viscous film entrained on a withdrawing plate A flat plate is withdrawn vertically upward from a large bath of viscous liquid at constant speed U under gravitational acceleration g (see figure 1). A thin film of liquid is entrained on the plate. Let x denote the vertical coordinate measured upward along the plate, and let $h(x)$ be the film thickness measured normal to the plate. Assume that the flow is steady and two-dimensional, liquid has viscosity μ , density ρ , and surface tension γ , the surrounding gas has negligible viscosity, the film is thin, so lubrication theory applies, far above the bath, the film thickness tends to a constant value h_∞ , and in the meniscus region, capillarity and viscous stresses are both important.

1. Starting from the Navier–Stokes equations, derive the lubrication equations for the film on the plate.
2. Show that the pressure in the liquid is independent of the coordinate across the film and is given by the capillary pressure

$$p = p_{\text{atm}} - \gamma h''(x),$$

to leading order.

3. Solve for the velocity profile $u(y)$ in the film, using no-slip on the plate and zero tangential stress at the free surface.
4. Derive the volume flux per unit width,

$$q = \int_0^h u \, dy,$$

and show that

$$q = Uh - \frac{\gamma}{3\mu} h^3 h''.$$

5. Explain why $q = Uh_\infty$, and hence derive the equation governing $h(x)$,

$$Uh_\infty = Uh - \frac{\gamma}{3\mu} h^3 h''.$$

6. Introduce dimensionless variables by scaling h and x with h_∞ and a characteristic meniscus lengthscale l , and show that the problem is governed by the capillary number

$$\text{Ca} = \frac{\mu U}{\gamma},$$

and the aspect ratio h_∞/l .

7. Use this result to obtain the pressure gradient $\partial p/\partial x$ as a function of h , γ , and the capillary number.

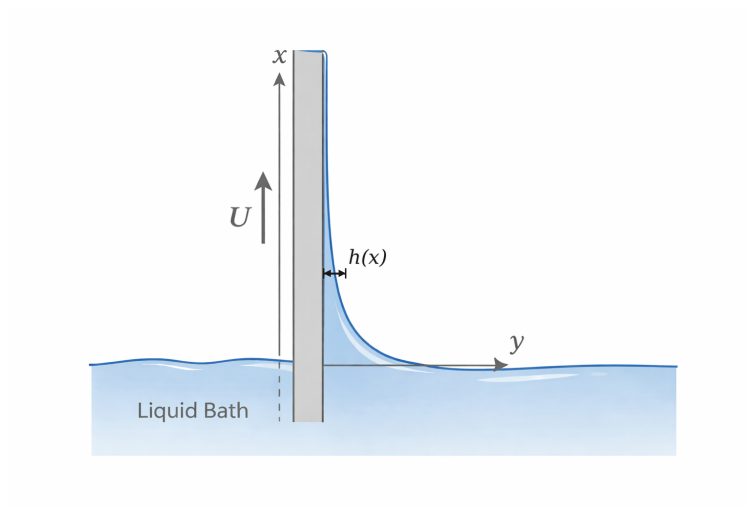


Figure 1: Entrainment by raising plate.